

Week -1 Requirement Analysis and Customer Database Module

1.Requirement Analysis

Understanding Business for Beauty Products

The Beauty Products Sales System manages the end-to-end flow of discovering, purchasing, and supplying cosmetics, skincare, haircare, and wellness products online. Below is a structured breakdown of the business process flow, entity definitions, and data requirements for system design and database modeling.

Process Flow:

[Browse / Search Product] → [Add to Cart] → [Order Placed] → [Checkout/ Payment] → [Delivery] → [Review & Support]

Browse / Search Product:

The customer explores beauty products by category (e.g., Skincare, Makeup, Haircare, Fragrance) or searches by brand, skin type, or ingredient preference.

Add to Cart:

After choosing the required items, the customer adds them to their shopping cart and selects specific shades or quantity variants.

Order Placed:

The customer provides or selects a shipping address and places the order.

Checkout / Payment:

The system directs the user to a secure payment gateway to process payments via UPI, Cards, or Net Banking.

Delivery:

Once payment is processed, the order is assigned to the seller/warehouse for dispatch and delivery to the customer.

Review & Support:

Following delivery, customers leave product ratings and reviews, or request assistance regarding returns, replacements, or beauty product inquiries.

Entity	Description	Key Attributes	Key Relationships
Customer	End-user who browses and purchases beauty products.	customer_id (PK), first_name, last_name, email, phone, created_at	One-to-Many with Order, Review
Seller	Brand, vendor, or store listing beauty products on the platform.	seller_id (PK), company_name, email, phone, rating, bank_details	One-to-Many with Product
Product	Beauty item listed for sale on the platform.	product_id (PK), seller_id (FK), title, description, price, stock_quantity, sku, category	Many-to-One with Seller, One-to-Many with Review
Order	Transaction instance representing a customer's beauty purchase.	order_id (PK), customer_id (FK), order_date, status, total_amount, shipping_address	Many-to-One with Customer, One-to-One with Payment, Many-to-Many with Product
Payment	Monetary transaction tied directly to an order.	payment_id (PK), order_id (FK), payment_method, status, transaction_id, amount, payment_date	One-to-One with Order
Review	Feedback and rating submitted by a customer for a beauty product.	review_id (PK), customer_id (FK), product_id (FK), rating, comment, created_at	Many-to-One with Customer, Many-to-One with Product

Customer table Creation:

```
SQL>
SQL> CREATE TABLE customer (
2     customer_id VARCHAR2(20) PRIMARY KEY,
3     first_name VARCHAR2(20),
4     last_name VARCHAR2(20),
5     email VARCHAR2(30),
6     phone NUMBER(10),
7     created_at DATE
8 );
```

Table created.

```
SQL> desc customer
```

Name	Null?	Type
CUSTOMER_ID	NOT NULL	VARCHAR2(20)
FIRST_NAME		VARCHAR2(20)
LAST_NAME		VARCHAR2(20)
EMAIL		VARCHAR2(30)
PHONE		NUMBER(10)
CREATED_AT		DATE

```
SQL> |
```

Insert values,using substitution variables:

```
SQL> INSERT INTO customer VALUES ('C-1001', 'Ananya', 'Sharma', 'ananya.s@glowbeauty.com', 9876543210, TO_DATE('2026-01-10', 'YYYY-MM-DD'));
1 row created.

SQL>
SQL> INSERT INTO customer VALUES ('C-1002', 'Priya', 'Rajan', 'priya.r@glowbeauty.com', 9876543211, TO_DATE('2026-02-15', 'YYYY-MM-DD'));
1 row created.

SQL>
SQL> INSERT INTO customer VALUES ('C-1003', 'Diya', 'Kapoor', 'diya.k@glowbeauty.com', 9876543212, TO_DATE('2026-03-05', 'YYYY-MM-DD'));
1 row created.

SQL>
SQL> INSERT INTO customer VALUES ('C-1004', 'Kavya', 'Sundar', 'kavya.s@glowbeauty.com', 9876543213, TO_DATE('2026-04-12', 'YYYY-MM-DD'));
1 row created.

SQL>
SQL> INSERT INTO customer VALUES ('C-1005', 'Meera', 'Nair', 'meera.n@glowbeauty.com', 9876543214, TO_DATE('2026-05-20', 'YYYY-MM-DD'));
1 row created.

SQL>
SQL> COMMIT;|
```

Like this we can create many rows we want as it.

Viewing table :

```
SQL> SELECT * FROM customer;
```

CUSTOMER_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	CREATED_A
C-1001	Ananya	Sharma	ananya.s@glowbeauty.com	9876543210	10-JAN-26
C-1002	Priya	Rajan	priya.r@glowbeauty.com	9876543211	15-FEB-26
C-1003	Diya	Kapoor	diya.k@glowbeauty.com	9876543212	05-MAR-26
C-1004	Kavya	Sundar	kavya.s@glowbeauty.com	9876543213	12-APR-26
C-1005	Meera	Nair	meera.n@glowbeauty.com	9876543214	20-MAY-26

Update:

```
SQL> UPDATE customer
  2 SET phone = 9870654312
  3 WHERE customer_id = 'C-1004';
```

1 row updated.

```
SQL> UPDATE customer
  2 SET email = 'ananya.new@glowbeauty.com',
  3      last_name = 'Verma'
  4 WHERE customer_id = 'C-1001';
```

1 row updated.

```
SQL> |
```